Linear Regression: Predicting NYC Airbnb Prices

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Dataset source: New York City Airbnb Open Data (AB_NYC_2019), Kaggle. The listings are from 2019.

1. Objective

Use a simple linear regression model to estimate a New York City Airbnb listing's nightly price from its location, room type, review activity, and availability. This project follows the course workflow and is a baseline prediction, not a current pricing tool.

2. Dataset Overview

The dataset contains **48,895 listings and 16 columns**. The target is price in US dollars per night. Model inputs include latitude/longitude, minimum nights, review counts, availability, borough, and room type. After cleaning and trimming, **45,759 listings** remained. The data was split into 80% training and 20% testing (random_state=42).

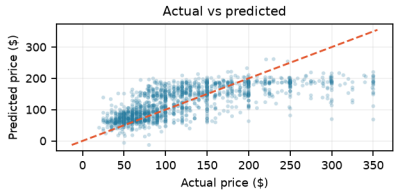
3. Data Quality and Cleaning

- **Missing values:** the 10,052 blank reviews-per-month values were filled with 0; last-review date and ID/name columns were dropped.
- **Outliers:** listings above the 95th-percentile price (**$355**) or minimum nights (**30**) were removed to focus on typical stays.
- **Features:** the specific neighbourhood label was dropped; borough and room type were converted to indicator columns. The final model used 13 features.

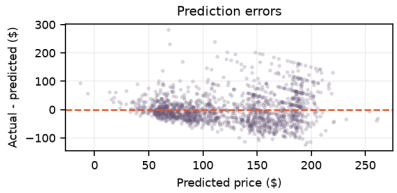
4. Model Results

Measure	Result	Meaning
R-squared	0.471 (47.1%)	The model explains about 47% of price differences in the test listings.
Mean Absolute Error (MAE)	$38.54	Predictions are off by about $39 per night on average.
Root Mean Squared Error (RMSE)	$53.24	Larger prediction mistakes count more heavily than with MAE.

5. Visual Results



Actual prices vs. predictions



Prediction errors



Error by price group

6. Key Findings and Limitations

- **Cheaper listings were easier to predict:** average absolute error was $29.03 for listings at or below the test-set median ($100) and $48.58 above it.
- The model captures some useful patterns, but about half of price variation remains unexplained. Special amenities, exact neighbourhood, and other factors are not represented.
- These results use 2019 listings and exclude the highest 5% by price and minimum nights; they do not describe all listings or today's prices.

7. Conclusion

The model provides a reasonable classroom baseline: its average test prediction is about $39 per night away from the listed price, and it performs better on cheaper listings. Adding more useful listing details or trying a more flexible model could improve estimates.

Source: Kaggle, “New York City Airbnb Open Data” by AB NYC. Model: scikit-learn LinearRegression; 80/20 holdout split, random_state=42.